

Supporting Information

Interpretable Machine Learning with Physically Informed Feature Engineering for Modeling the Simulated Performance of A_3BX_3 -Type Perovskite Solar Cells

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Contents

1. S1. Complete input simulation parameters for all device structures
2. S2. Original descriptors and target variables
3. S3. Complete definitions of physically derived features
4. S4. Data cleaning, preprocessing, and model settings
5. S5. Complete comparison of photovoltaic parameters before and after optimization
6. S6. Structure-level leave-one-out validation results
7. S7. Additional one-factor parameter-effect curves
8. S8. Notes on model interpretation

S1 Complete Input Simulation Parameters for All Device Structures

The main text lists only representative absorber-layer parameters to keep the manuscript concise. The complete input simulation parameters of the absorber, electron transport layer (ETL), and hole transport layer (HTL) used in the machine-learning dataset are provided in Table S1. To improve readability, Table S1 is divided into four consecutive parts covering Structures 1–8, 9–16, 17–24, and 25–31. The structure labels are used only in the Supporting Information to keep the wide parameter tables compact; the main text reports representative optimized devices using their full layer sequences. The listed parameters include layer thickness, band gap, electron affinity, relative dielectric constant, carrier mobility, doping concentration, and defect density. These parameters were used to construct the original descriptors for the machine-learning models.

Table S1: Input simulation parameters of absorber, ETL, and HTL layers for Structures 1–8.

Parameters	Structure 1	Structure 2	Structure 3	Structure 4	Structure 5	Structure 6	Structure 7	Structure 8
Absorber layer								
Absorber thickness (μm)	0.05–1.25	0.1–1.4	0.05–1.5	0.1–1.2	0.05–1.2	0.05–1	0.05–1.2	0.1–1.3
Absorber band gap (eV)	1.309	1.25	1.522	1.75	1.2	0.997	2.01	1.258
Electron affinity (eV)	4.181	4.198	4.22	3.97	4	4.5	3.78	3.85
Dielectric permittivity	5.93	5.85	4.78	4.95	5.32	6.484	7.729	5.9
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	100	25	50	50	50	100	20
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	20	20	50	50	50	20	25
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{18}	10^{14} – 10^{17}	10^{15} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{20}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Electron transport layer								
ETL thickness (μm)	0.05–0.5	0.05–0.5	0.05–0.45	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.6
ETL band gap (eV)	2.4	2	2	3.26	3.05	1.8–1.85	1.8	1.85
Electron affinity (eV)	4.2	3.9	3.9	4.26	4.16	3.9–4.26	3.9	4.26
Dielectric permittivity	10	3.9	3.9	32	10	13.6–17.7	13.6	17.7
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	100	0.2	0.2	20	15	50–100	100	50
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	25	0.2	0.2	10	0.1	25–100	100	25
Donor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Hole transport layer								
HTL thickness (μm)	0.05–0.5	0.1–0.4	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.45	0.05–0.5	0.05–0.5
HTL band gap (eV)	3.3	1.9	1.9	2.17	2.2	1.9	2.98	2.2
Electron affinity (eV)	2.2	3.6	3.6	3	3.4	3.6	2.1	3.4
Dielectric permittivity	9.77	5.4	5.4	7.5	8	5.4	6.5	7.5
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	0.1	30	30	200	100	30	1.69×10^{-4}	200
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	0.46	10	10	80	150	10	1.69×10^{-4}	8600
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Refs	[1, 2, 3]	[4, 5, 6]	[5, 6, 7]	[8, 9]	[9, 10, 11]	[12, 13, 6]	[4, 13, 6]	[14, 15, 5]

Table S1. Continued. Input simulation parameters of absorber, ETL, and HTL layers for Structures 9–16.

Parameters	Structure 9	Structure 10	Structure 11	Structure 12	Structure 13	Structure 14	Structure 15	Structure 16
Absorber layer								
Absorber thickness (μm)	0.1–1.2	0.05–1.2	0.1–1.2	0.1–1.2	0.1–1.2	0.1–1.2	0.1–1.2	0.1–1.2
Absorber band gap (eV)	1.077	1.25	1.43	0.82	1.598	1.269	1.307	1.577
Electron affinity (eV)	4	4.198	4.1	4	4.153	4.17	4	3.89
Dielectric permittivity	7	5.85	4.8	7.7	5.8	5.3	5.4	5.65
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	100	100	100	50	20	50	50
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	20	20	20	50	10	50	25
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Electron transport layer								
ETL thickness (μm)	0.05–0.5	0.05–0.5	0.05–0.45	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.5
ETL band gap (eV)	1.7	1.7	1.85–2.42	1.7	2.9–3.12	3.2	3.03	3.3
Electron affinity (eV)	3.91	3.9	4.26–4.3	3.91	4.09–4.4	4	3.6	4
Dielectric permittivity	4.2	4.2	9.35–17.7	4.2	10–22	8.7	9	9
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	0.08	0.08	50–100	0.08	0.69–50	5300	100	150
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	0.004	0.004	25	0.004	0.69–20	660	25	25
Donor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Hole transport layer								
HTL thickness (μm)	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.2	0.05–0.5	0.05–0.5	0.05–0.5
HTL band gap (eV)	1.4	1.5	1.04–1.9	1.12	1.04–2.98	1.34	3	1.58
Electron affinity (eV)	3.9	3.55	3.6–4.5	4.17	2.1–4.3	4.14	2.2	4.2
Dielectric permittivity	13.6	13.6	5.4–13.6	11.9	6.5–12	10	3	14.6
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	100	25	30–100	1500	1.69×10^{-4} –500	100	0.002	49
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	25	25	10–25	480	1.69×10^{-4} –25	25	0.002	49
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}
Refs	[16, 5, 2]	[4, 13, 17]	[18, 19, 2]	[4, 5, 20]	[1, 21, 2]	[13]	[1]	[22]

Table S1. Continued. Input simulation parameters of absorber, ETL, and HTL layers for Structures 17–24.

Parameters	Structure 17	Structure 18	Structure 19	Structure 20	Structure 21	Structure 22	Structure 23	Structure 24
Absorber layer								
Absorber thickness (μm)	0.7–1.2	0.1–1.2	0.1–1	0.4–1.2	0.1–1.2	0.1–1.2	1	0.2–1.2
Absorber band gap (eV)	0.78	0.952	0.952	0.976	1.307	1.309	1.528	1.528
Electron affinity (eV)	4	4.311	4.25	4.554	4	4.181	4.16	4.16
Dielectric permittivity	7	12.111	6.606	13.634	5.4	5.93	5.28	5.28
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	50	50	50	50	50	25	25
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	50	50	50	50	50	20	20
Acceptor density (cm^{-3})	10^{14} – 10^{16}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{16} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}
Electron transport layer								
ETL thickness (μm)	0.05–0.5	0.05–0.5	0.05–0.25	0.05–0.5	0.05–0.5	0.05–0.5	0.03–0.1	0.05–0.5
ETL band gap (eV)	2.24	2.8	2.1	1.85	2.24	2.42	2.1	2.24
Electron affinity (eV)	4.24	3.8	4.65	4.26	4.24	4.3	4.65	4.24
Dielectric permittivity	10	9	13.5	17.7	10	9.35	13.5	10
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	100	400	50	50	100	400	50
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	50	25	210	25	50	25	210	50
Donor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{16}	10^{14} – 10^{15}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}	10^{14} – 10^{19}
Hole transport layer								
HTL thickness (μm)	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.5	0.05–0.4	0.05–0.5	0.1	0.05–0.5
HTL band gap (eV)	3.6	1.4	3	3.1	2.2	3.4	2.2	3.8
Electron affinity (eV)	1.7	3.8	2.5	2.1	3.4	2.1	3.4	1.46
Dielectric permittivity	10	9	12.5	6.5	8	10	8	11.7
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	100	100	25	100	150	100	100	2.8
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	25	12.5	100	43.9	100	25	150	12
Acceptor density (cm^{-3})	10^{16} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Refs	[16, 5, 2]	[4, 13, 17]	[18, 19, 2]	[4, 5, 20]	[1, 21, 2]	[13]	[1]	[23]

Table S1. Continued. Input simulation parameters of absorber, ETL, and HTL layers for Structures 25–31.

Parameters	Structure 25	Structure 26	Structure 27	Structure 28	Structure 29	Structure 30	Structure 31
Absorber layer							
Absorber thickness (μm)	1	1–1.2	0.1–1.2	0.1–1.2	0.1–1.2	0.1–1.2	0.1–1.2
Absorber band gap (eV)	1.649	1.93	1.957	1.265	1.164	1.635	1.269
Electron affinity (eV)	4.12	4.19	4.13	3.92	4.22	3.85	4.17
Dielectric permittivity	5.75	4.73	4.98	5.85	14.24	18.44	5.3
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	80	50	50	25	80	80	20
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	20	25	25	20	25	25	10
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{18}	10^{14} – 10^{18}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}
Electron transport layer							
ETL thickness (μm)	0.05–0.5	0.15	0.05–0.5	0.05–0.5	0.05–0.3	0.05–0.3	0.1–0.5
ETL band gap (eV)	3.3	3.3	3.5	2.8	2.9	2	1.8
Electron affinity (eV)	4	4	4	3.8	4.09	3.9	3.9
Dielectric permittivity	66	66	9	9	10	3.9	13.6
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	100	100	20	100	50	0.2	100
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	25	25	10	25	20	0.2	100
Donor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Defect density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}
Hole transport layer							
HTL thickness (μm)	0.05–0.5	0.05	0.05–0.5	0.05–0.3	0.05–0.3	0.05–0.3	0.1–0.5
HTL band gap (eV)	1.5	1.72	1.55	1.55	2.96	1.55	2.2
Electron affinity (eV)	4.2	4.09	4.03	3.6	2.3	3.6	3.4
Dielectric permittivity	7.11	5.78	13.8	6.5	9	6.5	8
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	1	600	30	12	1	12	150
Hole mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	3.8	200	30	9	40	9	100
Acceptor density (cm^{-3})	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{17}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{19} – 10^{20}
Defect density (cm^{-3})	10^{14} – 10^{15}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14} – 10^{19}	10^{14}
Refs	[24]	[23]	[23]	[23]	[25]	[26]	[13]

Note: E_g , band gap; χ , electron affinity; ϵ_r , relative dielectric constant; μ_n , electron mobility; μ_p , hole mobility; N_A , acceptor density; N_D , donor density; N_t , defect density. For machine-learning modeling, thicknesses were converted to nm, while doping concentration and defect density were expressed in logarithmic form where appropriate. The full layer sequence of each numbered structure is listed in Table S5.

S2 Original Descriptors and Target Variables

The original descriptor set contains 24 input variables from the absorber, ETL, and HTL, together with four photovoltaic target variables. The descriptor definitions are listed in Table S2. These descriptors describe layer thickness, band gap, electron affinity, relative dielectric constant, carrier mobility, doping concentration, and defect density.

Table S2: Original descriptors and target variables used in the machine-learning models.

Variable name	Layer / type	Unit	Physical meaning
Abs_Thickness_nm	Absorber	nm	Absorber thickness; affects optical absorption and carrier transport distance.
Abs_Eg_eV	Absorber	eV	Absorber band gap; controls absorption threshold and voltage–current trade-off.
Abs_Chi_eV	Absorber	eV	Absorber electron affinity; determines conduction-band position.

Variable name	Layer / type	Unit	Physical meaning
Abs_Eps_r	Absorber	–	Relative dielectric constant of absorber.
Abs_mu_n	Absorber	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Electron mobility in absorber.
Abs_mu_p	Absorber	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Hole mobility in absorber.
Abs_log10_NA	Absorber	$\log_{10}(\text{cm}^{-3})$	Logarithmic acceptor density in absorber.
Abs_log10_Nt	Absorber	$\log_{10}(\text{cm}^{-3})$	Logarithmic defect density in absorber.
ETL_Thickness_nm	ETL	nm	ETL thickness; affects electron transport path and resistive loss.
ETL_Eg_eV	ETL	eV	ETL band gap.
ETL_Chi_eV	ETL	eV	ETL electron affinity; affects electron extraction.
ETL_Eps_r	ETL	–	Relative dielectric constant of ETL.
ETL_mu_n	ETL	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Electron mobility in ETL.
ETL_mu_p	ETL	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Hole mobility in ETL.
ETL_log10_ND	ETL	$\log_{10}(\text{cm}^{-3})$	Logarithmic donor density in ETL.
ETL_log10_Nt	ETL	$\log_{10}(\text{cm}^{-3})$	Logarithmic defect density in ETL.
HTL_Thickness_nm	HTL	nm	HTL thickness; affects hole transport path and resistive loss.
HTL_Eg_eV	HTL	eV	HTL band gap.
HTL_Chi_eV	HTL	eV	HTL electron affinity; affects interfacial alignment.
HTL_Eps_r	HTL	–	Relative dielectric constant of HTL.
HTL_mu_n	HTL	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Electron mobility in HTL.
HTL_mu_p	HTL	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	Hole mobility in HTL.
HTL_log10_NA	HTL	$\log_{10}(\text{cm}^{-3})$	Logarithmic acceptor density in HTL.
HTL_log10_Nt	HTL	$\log_{10}(\text{cm}^{-3})$	Logarithmic defect density in HTL.
V_{oc}	Target	V	Open-circuit voltage.
J_{sc}	Target	mA cm^{-2}	Short-circuit current density.
FF	Target	%	Fill factor.
PCE	Target	%	Power conversion efficiency.

S3 Complete Definitions of Physically Derived Features

The complete definitions of physically derived features are listed in Table S3. These features were constructed to explicitly describe transport balance, interfacial energy-level alignment, band-gap matching, defect distribution, thickness–defect coupling, and transport bottlenecks.

Table S3: Complete definitions of physically derived features used in the machine-learning models.

Feature group	Derived feature	Definition	Physical meaning
Mobility-ratio features	Abs_mu_ratio_log	$\log_{10}(\mu_{n,Abs}/\mu_{p,Abs})$	Transport balance in absorber.
Mobility-ratio features	ETL_mu_ratio_log	$\log_{10}(\mu_{n,ETL}/\mu_{p,ETL})$	Carrier selectivity of ETL.
Mobility-ratio features	HTL_mu_ratio_log	$\log_{10}(\mu_{n,HTL}/\mu_{p,HTL})$	Carrier selectivity of HTL.
Transport-matching features	ETL_HTL_transport_balance_log	$\log_{10}(\mu_{n,ETL}/\mu_{p,HTL})$	ETL/HTL transport matching.
Transport-bottleneck feature	Transport_mu_min_log	$\log_{10}[\min(\mu_{n,ETL}, \mu_{p,HTL})]$	Slow-side transport limitation.
Energy-level offset features	ETL_Abs_Chi_offset	$\chi_{ETL} - \chi_{Abs}$	ETL/absorber alignment.
Energy-level offset features	Abs_HTL_Chi_offset	$\chi_{Abs} - \chi_{HTL}$	Absorber/HTL alignment.
Band-gap offset features	ETL_Abs_Eg_offset	$E_{g,ETL} - E_{g,Abs}$	ETL/absorber band-gap matching.
Band-gap offset features	HTL_Abs_Eg_offset	$E_{g,HTL} - E_{g,Abs}$	HTL/absorber band-gap matching.
Defect-related features	Total_log10_Nt	$\log_{10} N_{t,Abs} + \log_{10} N_{t,ETL} + \log_{10} N_{t,HTL}$	Overall defect burden.
Defect-difference features	Abs_Nt_minus_ETL_Nt	$\log_{10} N_{t,Abs} - \log_{10} N_{t,ETL}$	Absorber/ETL defect difference.
Defect-difference features	Abs_Nt_minus_HTL_Nt	$\log_{10} N_{t,Abs} - \log_{10} N_{t,HTL}$	Absorber/HTL defect difference.
Defect-difference features	ETL_Nt_minus_HTL_Nt	$\log_{10} N_{t,ETL} - \log_{10} N_{t,HTL}$	Transport-layer defect asymmetry.
Thickness-defect coupling features	Abs_thickness_x_Abs_Nt	$d_{Abs} \times \log_{10} N_{t,Abs}$	Absorber recombination-length coupling.
Thickness-defect coupling features	ETL_thickness_x_ETL_Nt	$d_{ETL} \times \log_{10} N_{t,ETL}$	ETL defect-loss coupling.
Thickness-defect coupling features	HTL_thickness_x_HTL_Nt	$d_{HTL} \times \log_{10} N_{t,HTL}$	HTL defect-loss coupling.
Interfacial-mismatch features	Abs_ETL_Chi_mismatch_abs	$ \chi_{ETL} - \chi_{Abs} $	ETL/absorber mismatch.
Interfacial-mismatch features	Abs_HTL_Chi_mismatch_abs	$ \chi_{Abs} - \chi_{HTL} $	Absorber/HTL mismatch.

S4 Data Cleaning, Preprocessing, and Model Settings

The additional preprocessing and modeling details are summarized in Table S4. The preprocessing workflow was designed to reduce format-related errors and physically inconsistent records while avoiding data leakage.

Table S4: Summary of data preprocessing and model-validation settings.

Item	Setting	Purpose
Numerical standardization	Unit symbols, spaces, and non-standard numerical characters were converted to numerical values.	To ensure all descriptors could be used for model training.
Missing-value treatment	Samples with missing input descriptors or target values were removed.	To avoid incomplete records.
PCE consistency check	PCE was recalculated from V_{oc} , J_{sc} , and FF. Samples with absolute deviation larger than 0.5 were removed.	To reduce physically inconsistent labels.
Train-only IQR filtering	Outliers were identified using $[Q_1 - 3IQR, Q_3 + 3IQR]$ within the training set only.	To avoid information leakage from the test set.
Data split	Training/test ratio of 80:20.	To evaluate independent-test generalization.
Cross-validation	Five-fold stratified cross-validation based on target-value bins.	To reduce the influence of uneven target distributions.
Robustness test	Repeated random-seed evaluation.	To examine model stability under different data splits.
Structure-level leave-one-out validation	Each complete device structure was held out once as the test set, and the remaining structures were used for model training.	To evaluate extrapolation behavior for unseen device stacks.
ML-guided SCAPS-1D optimization	Machine-learning interpretation results were used to guide SCAPS-1D parameter optimization.	To connect model interpretation with practical device-design trends.
Models	RF, XGBoost, MLP, and DeepMLP.	To compare tree-based and neural-network-based regression models.

S5 Complete Comparison of Photovoltaic Parameters Before and After Optimization

The main text lists the top five optimized device structures ranked by PCE. The complete comparison of photovoltaic parameters before and after optimization for all device structures is given in Table S5. The optimized device structures are written using their full layer sequences to maintain consistency with the main text.

Table S5: Comparison of photovoltaic parameters before and after optimization for different lead-free perovskite solar-cell device structures.

No.	Device structure	Before optimization				After optimization			
		V_{OC} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)	V_{OC} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)
1	FTO/CdS/Sr ₃ SbI ₃ /SrCu ₂ O ₂ /Au	0.9642	34.0591	87.15	28.62	1.0781	34.5177	88.82	33.06
2	FTO/PCBM/Ca ₃ NBr ₃ /CBTS/Au	0.9881	20.2556	85.27	17.07	0.9944	36.2428	87.73	31.62
3	FTO/PCBM/Sr ₃ AsBr ₃ /CBTS/Au	1.1612	26.7917	88.60	27.57	1.2518	26.3247	87.12	28.71
4	FTO/TiO ₂ /Sr ₃ NCl ₃ /Cu ₂ O/Au	1.3253	15.8630	78.10	16.42	1.2824	17.2156	83.71	18.48
5	FTO/IGZO/Ba ₃ NCl ₃ /V ₂ O ₅ /Au	0.9320	38.1684	87.37	31.08	0.9887	38.2986	88.02	33.33
6	FTO/WS ₂ /Ba ₃ PCl ₃ /CBTS/Au	0.6859	6.77655	66.58	3.09	1.0891	16.5225	86.44	15.56
7	FTO/WS ₂ /Ca ₃ NF ₃ /CuI/Au	1.2783	12.0993	76.43	11.82	1.2049	16.0521	78.98	15.28
8	FTO/SnS ₂ /Sr ₃ PI ₃ /Cu ₂ O/Au	0.9203	33.9029	78.08	24.36	0.9314	36.0269	81.00	27.18
9	FTO/C ₆₀ /Ca ₃ NI ₃ /CIGSe/Au	0.7071	41.3671	83.64	24.46	0.8261	43.1267	86.10	30.67
10	FTO/C ₆₀ /Ca ₃ NBr ₃ /CIS/Au	0.9599	25.0688	81.54	19.62	0.9714	36.0149	86.46	30.25
11	FTO/CdS/Ca ₃ NCl ₃ /CISe/Au	0.9478	23.5329	83.69	18.67	1.0327	26.6837	87.79	24.19
12	FTO/C ₆₀ /Ca ₃ NI ₃ /Si/Au	0.4606	50.2524	78.17	18.09	0.5194	52.1862	80.54	21.83
13	FTO/LBSO/Ca ₃ SbI ₃ /CISse/Au	0.8086	14.1006	53.92	6.15	1.1251	13.0751	74.15	10.91
14	FTO/STO/Mg ₃ AsBr ₃ /CTSe/Au	0.8908	33.4314	86.18	25.67	0.9202	35.7998	85.99	28.33
15	FTO/ZnO/Sr ₃ SbI ₃ /Spiro-MeOTAD/Au	1.0153	32.278973	87.39	28.64	0.9956	34.849045	87.79	30.46
16	FTO/ZnO/Ca ₃ AsI ₃ /CuSbS ₂ /Au	0.9977	25.193681	87.53	22.00	1.0231	25.696212	87.52	23.01
17	FTO/SnS ₂ /Ba ₃ SbI ₃ /CuSCN/Au	0.4047	8.30652	43.93	1.48	0.4169	8.442776	43.65	1.54
18	FTO/ZnS/Ba ₃ AsBr ₃ /CZTS/Au	0.6435	2.44925	49.14	0.77	0.6990	45.989537	82.20	26.43
19	FTO/SnS ₂ /Ba ₃ PBr ₃ /MoO ₃ /Au	0.7008	0.076986776	68.02	0.04	0.7085	0.076763535	68.06	0.04
20	FTO/SnS ₂ /Ba ₃ SbBr ₃ /CuI/Au	0.6322	47.997807	80.91	24.55	0.7290	47.929169	84.09	29.38
21	FTO/SnS ₂ /Sr ₃ SbI ₃ /V ₂ O ₅ /Au	1.0163	32.35723	87.83	28.88	1.0008	34.62154	87.97	30.48
22	FTO/CdS/Sr ₃ SbI ₃ /CuSCN/Au	0.9665	34.063388	87.21	28.71	1.0786	34.516652	88.65	33.00
23	FTO/In ₂ P ₃ /Sr ₃ PBr ₃ /V ₂ O ₅ /Au	1.1854	0.090827742	61.66	0.07	1.1020	0.10248648	61.65	0.07
24	FTO/SnS ₂ /Sr ₃ PBr ₃ /NiO/Au	1.0809	24.683146	83.71	22.33	1.1776	23.560392	89.37	24.80
25	FTO/MZO/Sr ₃ PCl ₃ /Zn ₂ P ₃ /Au	1.1491	23.816871	88.91	24.33	1.1536	24.081155	88.91	24.70
26	FTO/MZO/Ca ₃ AsBr ₃ /BiI ₃ /Au	1.2424	11.445746	76.39	10.86	1.1639	15.224548	84.38	14.95
27	FTO/SnO ₂ /Ca ₃ PBr ₃ /WSe ₂ /Au	1.3507	12.028964	62.96	10.23	1.1478	13.662805	84.72	13.29
28	FTO/ZnS/Sr ₃ AsI ₃ /CuS/Au	0.9135	35.686133	86.75	28.28	1.0280	35.878468	88.44	32.62
29	FTO/ZnSe/Sr ₃ BiI ₃ /PTAA/Au	0.8711	39.957026	84.91	29.55	0.9265	40.392377	86.95	32.54
30	FTO/PCBM/Ca ₃ BiCl ₃ /CuS/Au	1.1792	19.801253	88.74	20.72	1.1700	24.233693	88.42	25.07
31	FTO/WS ₂ /Mg ₃ AsBr ₃ /V ₂ O ₅ /Au	1.0760	18.44872	87.16	17.30	1.0929	33.11436	88.64	32.08

Note: $\Delta PCE = PCE_{after} - PCE_{before}$. The improvement percentage is calculated as $(PCE_{after} - PCE_{before}) / PCE_{before} \times 100\%$.

S6 Structure-Level Leave-One-Out Validation Results

To further evaluate whether the trained models can generalize beyond record-level random splitting, structure-level leave-one-out validation was performed. In each validation run, all records associated with one complete device structure were held out as the test set, and the remaining structures were used for model training. This setting is more stringent than the random 80:20 split because the model is required to predict an unseen device stack rather than unseen records from already represented structures.

Table S6 summarizes the structure-level leave-one-out validation results over all 31 held-out structures using the pooled out-of-fold R^2 , the mean $\pm$ standard deviation of RMSE and MAE, the median normalized

RMSE, and the number of structures with positive per-structure R^2 . Table S7 lists the complete per-structure results for V_{oc} , J_{sc} , FF, and PCE. Overall, the structure-level leave-one-out results are clearly more challenging than the random-split evaluation reported in the main text. Some held-out structures show positive R^2 values, whereas others show negative values, indicating that extrapolation to a completely unseen device stack remains difficult. Therefore, these results support a cautious interpretation: the present models are useful for preliminary screening and trend analysis within the represented A_3BX_3 -type perovskite device space, but additional data and physical validation are still required for reliable extrapolation to new device structures.

Table S6: Summary of structure-level leave-one-out validation results based on pooled out-of-fold predictions over Structures 1–31.

Target	Held-out structures	Pooled out-of-fold R^2	RMSE mean $\pm$ SD	MAE mean $\pm$ SD	Median nRMSE	Positive per-structure R^2
V_{oc}	31	0.6966	0.0942 ± 0.0552	0.0787 ± 0.0517	0.3491	17/31
J_{sc}	31	0.3129	8.8552 ± 5.6853	8.0226 ± 5.5728	0.6819	11/31
FF	31	-0.2158	11.6365 ± 7.1772	9.3788 ± 6.4613	0.7540	11/31
PCE	31	0.4866	6.2207 ± 3.3881	5.4565 ± 3.2914	0.5772	15/31

Note: The pooled out-of-fold R^2 was calculated by concatenating the held-out predictions from all 31 structure-level leave-one-out runs and evaluating them against the global target variance. RMSE and MAE are reported as the mean $\pm$ standard deviation over the 31 held-out structures. nRMSE denotes RMSE normalized by the global standard deviation of the corresponding target variable. The number of positive per-structure R^2 values was calculated from the individual held-out-structure results listed in Table S7.

The complete structure-level leave-one-out validation results are listed in Table S7. To keep the table readable, Table S7 reports the held-out structure code and absorber material, while the corresponding full device structures are provided in Table S5. In each LOMO run, all records belonging to one held-out structure were used as the test set, and the selected XGBoost model was trained using records from the remaining structures. This setting is more stringent than random splitting because the model is required to predict a complete device stack that is absent from the training data. Negative R^2 values indicate that extrapolation to some unseen device structures remains difficult, especially when the held-out structure has a narrow target range or differs strongly from the training structures. In Table S7, R^2 is calculated separately within each held-out structure and is therefore a per-structure R^2 , whereas the pooled out-of-fold R^2 reported in Table S6 is calculated after concatenating all held-out predictions across the 31 structures.

Table S7: Complete per-structure leave-one-out validation results for Structures 1–31 using the selected XGBoost models.

Held-out structure	Absorber	Target	N_{test}	R^2	RMSE	MAE
S1	Sr_3SbI_3	V_{oc}	69	0.8210	0.0314	0.0185
		J_{sc}	69	0.8847	1.8187	1.6806
		FF	69	-0.4615	4.3083	3.2734
		PCE	69	0.5784	3.7238	2.7559
S2	Ca_3NBr_3	V_{oc}	51	0.9325	0.0186	0.0133
		J_{sc}	51	-0.2513	6.6108	6.4386
		FF	51	0.7502	6.6746	3.8838
		PCE	51	0.8431	2.8868	2.5295
S3	Sr_3AsBr_3	V_{oc}	70	-3.0278	0.1685	0.1648
		J_{sc}	70	0.4215	3.0003	2.5028
		FF	70	0.8238	4.8711	2.3114
		PCE	70	0.9315	1.5731	1.1566
S4	Sr_3NCl_3	V_{oc}	67	-8.4592	0.1848	0.1811
		J_{sc}	67	-19.3366	6.6220	6.4500
		FF	67	-0.2977	9.7693	8.3198
		PCE	67	-3.7824	5.9264	5.6256

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Held-out structure	Absorber	Target	N_{test}	R^2	RMSE	MAE
S5	Ba_3NCl_3	V_{oc}	73	0.8592	0.0192	0.0169
		J_{sc}	73	0.5452	4.7962	4.5873
		FF	73	-26.6726	7.1655	6.4968
		PCE	73	0.5915	4.1068	3.6943
S6	Ba_3PCl_3	V_{oc}	57	0.6899	0.1031	0.0719
		J_{sc}	57	-3.7459	15.2012	13.2858
		FF	57	-2.9605	19.0368	15.4546
		PCE	57	-3.3498	12.3102	10.5625
S7	Ca_3NF_3	V_{oc}	71	-2.2823	0.1516	0.1061
		J_{sc}	71	-0.5778	3.7762	2.9182
		FF	71	-0.0415	9.0044	6.3073
		PCE	71	-0.6319	4.1456	3.4743
S8	Sr_3PI_3	V_{oc}	59	-20.9638	0.1419	0.1404
		J_{sc}	59	-1.3707	8.1613	7.4102
		FF	59	-21.7974	13.1516	11.5288
		PCE	59	-0.7435	5.5759	4.8034
S9	Ca_3NI_3	V_{oc}	60	-0.0482	0.0597	0.0573
		J_{sc}	60	-1.5865	10.8907	10.3575
		FF	60	-10.8198	22.8571	20.9516
		PCE	60	-3.3761	12.3403	11.6006
S10	Ca_3NBr_3	V_{oc}	239	-1.9581	0.1871	0.1263
		J_{sc}	239	0.3329	8.8348	7.1999
		FF	239	-1.0639	27.3495	16.5093
		PCE	239	0.1896	8.2954	5.4854
S11	Ca_3NCl_3	V_{oc}	96	-0.0130	0.1198	0.1044
		J_{sc}	96	0.2957	5.0698	4.4845
		FF	96	-3.5501	22.8235	20.6199
		PCE	96	0.6343	3.8542	3.2368
S12	Ca_3NI_3	V_{oc}	64	-27.1295	0.1890	0.1857
		J_{sc}	64	-8.3500	18.7796	18.5853
		FF	64	-1.5123	11.8192	11.5818
		PCE	64	-2.7363	6.7634	6.4839
S13	Ca_3SbI_3	V_{oc}	60	0.2906	0.1141	0.0721
		J_{sc}	60	-0.2391	4.2976	3.1372
		FF	60	-0.5693	17.0348	10.8695
		PCE	60	-1.1276	4.3556	3.0048
S14	Mg_3AsBr_3	V_{oc}	112	0.4891	0.0577	0.0355
		J_{sc}	112	-1.0572	8.7114	7.9802
		FF	112	-4.0791	11.0606	9.4174
		PCE	112	-1.6057	8.4828	7.8298
S15	Sr_3SbI_3	V_{oc}	66	0.6991	0.0435	0.0402
		J_{sc}	66	0.5093	5.4861	2.5016
		FF	66	0.1737	17.0058	9.0408
		PCE	66	0.5238	5.7147	3.6920
S16	Ca_3AsI_3	V_{oc}	58	-3.2002	0.1490	0.1375
		J_{sc}	58	-2.8197	5.4800	5.2868
		FF	58	0.7436	3.1913	2.2729
		PCE	58	0.3066	3.1436	2.9295
S17	Ba_3SbI_3	V_{oc}	48	-10.3675	0.1588	0.1566
		J_{sc}	48	-309.6664	29.7784	28.9549
		FF	48	-381.5600	32.6107	32.4406
		PCE	48	-2.44×10^3	17.5491	17.2645
S18	Ba_3AsBr_3	V_{oc}	95	0.6497	0.0620	0.0473
		J_{sc}	95	0.4085	14.2813	11.4198
		FF	95	0.1786	11.3874	8.4946
		PCE	95	0.5772	6.3377	4.6792
S19	Ba_3PBr_3	V_{oc}	31	0.6217	0.0563	0.0445
		J_{sc}	31	-5.30×10^7	15.4893	14.4966
		FF	31	-2.36×10^3	10.2167	9.1136
		PCE	31	-3.67×10^5	4.3182	3.8012

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Held-out structure	Absorber	Target	N_{test}	R^2	RMSE	MAE
S20	Ba_3SbBr_3	V_{oc}	41	0.6471	0.0534	0.0357
		J_{sc}	41	0.0276	11.7925	11.4483
		FF	41	0.2029	11.2506	7.3463
		PCE	41	0.6780	5.0937	4.4019
S21	Sr_3SbI_3	V_{oc}	84	0.7958	0.0747	0.0594
		J_{sc}	84	0.4128	7.3160	6.0501
		FF	84	-0.1919	10.5871	9.4494
		PCE	84	0.7873	4.5122	3.9449
S22	Sr_3SbI_3	V_{oc}	43	-1.0445	0.1057	0.1039
		J_{sc}	43	0.8948	1.8027	1.1548
		FF	43	-1.0903	3.5172	2.9001
		PCE	43	0.5633	4.0389	3.6503
S23	Sr_3PBr_3	V_{oc}	12	-5.5271	0.1260	0.1068
		J_{sc}	12	-2.97×10^6	10.9692	9.9231
		FF	12	-107.2046	8.3427	7.1510
		PCE	12	-6.14×10^6	12.2204	10.7955
S24	Sr_3PBr_3	V_{oc}	41	-0.7567	0.0328	0.0264
		J_{sc}	41	-6.9060	7.1395	6.8868
		FF	41	-36.1876	9.1743	8.1806
		PCE	41	-1.0306	3.7860	3.5315
S25	Sr_3PCl_3	V_{oc}	79	0.6518	0.0727	0.0584
		J_{sc}	79	-0.4782	5.8250	5.5643
		FF	79	0.8243	5.3134	4.4678
		PCE	79	0.2084	5.9104	5.4388
S26	Ca_3AsBr_3	V_{oc}	32	0.4788	0.0929	0.0533
		J_{sc}	32	-20.0375	8.6623	8.1917
		FF	32	0.7309	4.5098	3.6391
		PCE	32	-3.8086	5.9700	5.5049
S27	Ca_3PBr_3	V_{oc}	50	0.0507	0.1646	0.1158
		J_{sc}	50	-40.9988	9.4543	9.1888
		FF	50	-0.1806	14.1136	11.5464
		PCE	50	-6.3453	7.5507	7.0811
S28	Sr_3AsI_3	V_{oc}	53	0.9386	0.0157	0.0106
		J_{sc}	53	-2.1533	12.1378	9.6682
		FF	53	0.5405	4.0184	2.7720
		PCE	53	-0.1988	7.5069	7.1997
S29	Sr_3BiI_3	V_{oc}	50	0.2293	0.0545	0.0528
		J_{sc}	50	0.2414	8.4465	7.7195
		FF	50	0.2768	6.7347	5.4829
		PCE	50	0.7660	4.0010	3.4962
S30	Ca_3BiCl_3	V_{oc}	52	0.6828	0.0462	0.0374
		J_{sc}	52	-1.8427	3.3987	3.1118
		FF	52	-3.1586	11.1407	9.3494
		PCE	52	-1.2104	4.9316	4.5939
S31	Mg_3AsBr_3	V_{oc}	40	-0.2971	0.0661	0.0599
		J_{sc}	40	-1.0875	10.4808	10.1140
		FF	40	0.3278	10.6921	9.5687
		PCE	40	0.4603	5.9166	4.9035

Note: For each run, all samples belonging to the held-out structure were used as the test set, and all remaining structures were used for training. RMSE and MAE follow the units of the corresponding target variables: V for V_{oc} , mA cm^{-2} for J_{sc} , and % for FF and PCE.

Overall, the structure-level LOMO results show that the model performs reasonably for some held-out structures, such as S15, but not uniformly for all device stacks. This confirms that the present model is more reliable for interpolation within the represented device-parameter space than for unrestricted extrapolation to completely unseen structures.

S7 Predictive Uncertainty and Applicability-Domain Diagnostics

This section provides the complete numerical results for the two reliability diagnostics summarized in the main text: the calibration of the predictive uncertainty intervals (Table S8) and the applicability-domain correlation analysis (Table S9).

Table S8 reports the prediction-interval coverage probability (PICP) and mean prediction-interval width (MPIW) for the central 50%, 80%, and 90% intervals, obtained with quantile gradient boosting and with random-forest inter-tree variance. For the quantile-XGBoost model the empirical coverage closely tracks the nominal level at all three intervals (e.g., 0.43/0.75/0.89 for V_{oc} at the 0.50/0.80/0.90 levels), confirming good calibration, whereas the random-forest intervals are consistently over-covered (and therefore wider than necessary). MPIW values follow the units of each target.

Table S8: Calibration of the predictive uncertainty intervals on the independent test set: prediction-interval coverage probability (PICP) and mean prediction-interval width (MPIW) for the central 50%, 80%, and 90% intervals, for quantile gradient boosting (XGB) and random-forest inter-tree variance (RF).

Target	Nominal	Quantile XGB		RF variance	
		PICP	MPIW	PICP	MPIW
V_{oc}	0.50	0.432	0.027	0.881	0.051
V_{oc}	0.80	0.746	0.071	0.958	0.096
V_{oc}	0.90	0.891	0.140	0.975	0.123
J_{sc}	0.50	0.467	2.08	0.879	3.65
J_{sc}	0.80	0.746	5.30	0.975	6.94
J_{sc}	0.90	0.901	11.71	0.993	8.91
FF	0.50	0.452	4.42	0.857	5.37
FF	0.80	0.709	8.42	0.960	10.21
FF	0.90	0.862	13.50	0.985	13.11
PCE	0.50	0.449	2.03	0.894	3.08
PCE	0.80	0.738	5.64	0.980	5.85
PCE	0.90	0.889	10.92	0.983	7.51

Table S9 reports the Spearman rank correlation between the per-structure extrapolation error (normalized RMSE over the 31 leave-one-out folds) and three domain-shift indicators: the unweighted k -NN distance, the importance-weighted k -NN distance, and the hard extrapolation fraction. Only the importance-weighted distance yields a consistently positive and (for FF) statistically significant association with error, supporting its use as a pre-screening applicability-domain flag.

Table S9: Spearman rank correlation ρ (with p -value) between per-structure normalized RMSE and three domain-shift indicators, over the 31 structure-level leave-one-out folds.

Target	Unweighted ρ (p)	Importance-weighted ρ (p)	Extrapolation ρ (p)
V_{oc}	0.01 (0.976)	0.15 (0.426)	0.02 (0.897)
J_{sc}	0.12 (0.507)	0.27 (0.142)	−0.10 (0.593)
FF	0.32 (0.078)	0.51 (0.004)	0.09 (0.639)
PCE	0.16 (0.404)	0.35 (0.054)	−0.02 (0.921)

S8 Additional One-Factor Parameter-Effect Curves

The main text presents representative one-factor parameter-effect curves. Additional parameter-effect curves are provided in Figures S1–S8. For each curve, one selected descriptor was varied within its data range, while the remaining descriptors were fixed at their median values in the training set. These curves should be interpreted as model-learned response trends rather than strict experimental single-variable effects.

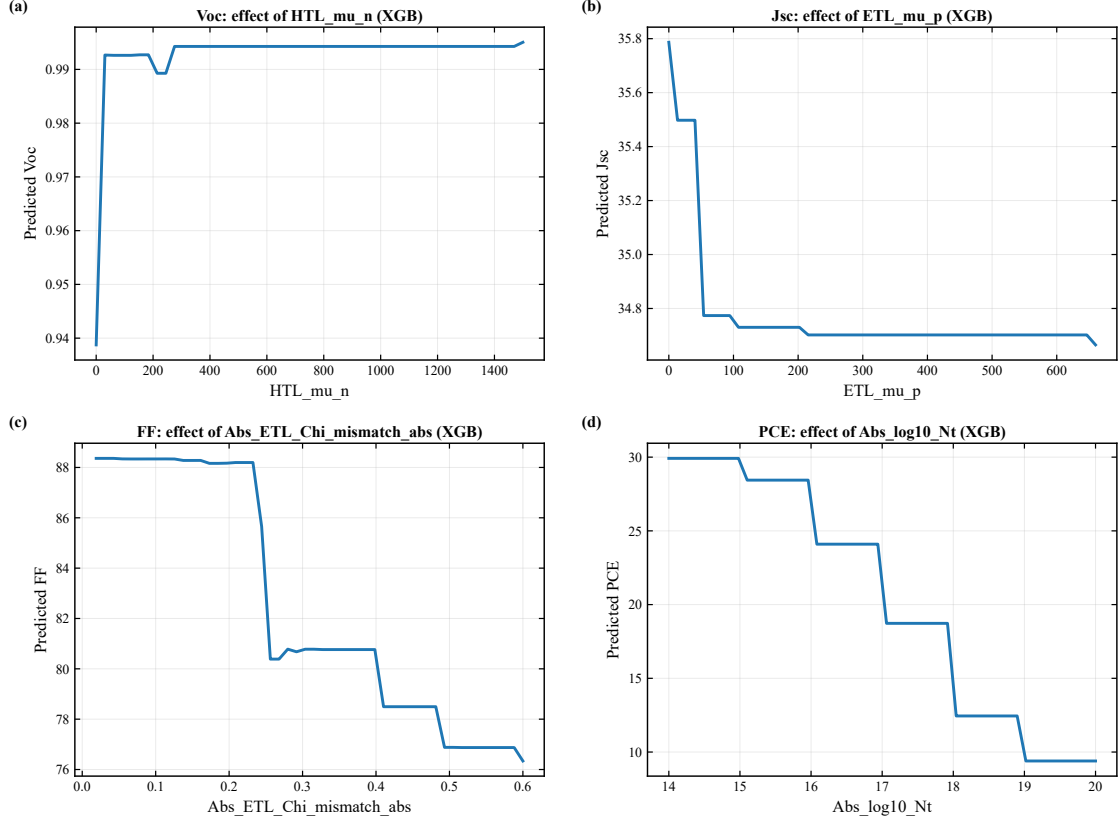


Figure S1: Additional one-factor parameter-effect curves, group 1.

S9 Notes on Model Interpretation

The model-interpretation results reported in the main text and Supporting Information reflect statistical relationships learned from the available dataset. They should not be interpreted as direct experimental causality. For one-factor parameter-effect analysis, only one descriptor was varied at a time, while the remaining descriptors were fixed at their median values. Because photovoltaic device parameters are physically coupled, these curves mainly provide qualitative trends learned by the XGBoost models.

Further experimental validation is required before the identified parameter windows can be used as definitive material-design rules. Nevertheless, the additional results provided here support the main conclusion that absorber defect density, absorber band gap, interfacial energy-level mismatch, and carrier-transport balance are key factors influencing the performance of A_3BX_3 -type perovskite solar cells.

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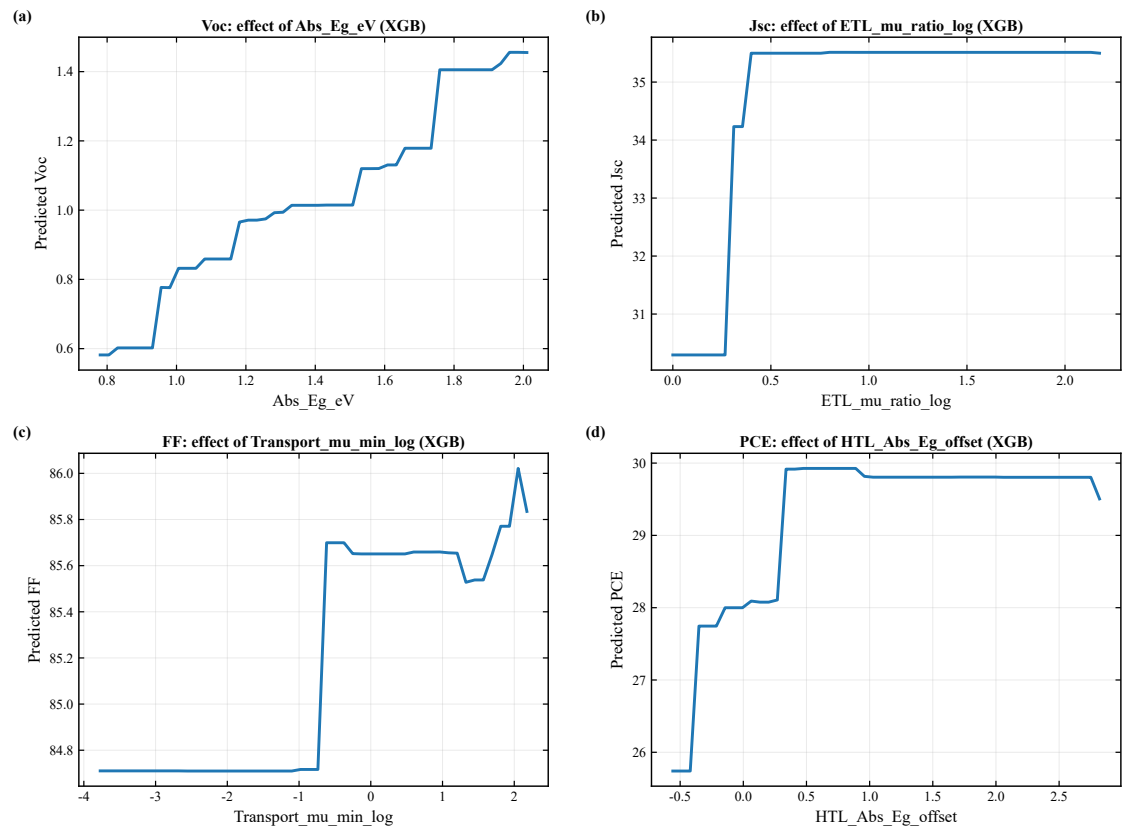


Figure S2: Additional one-factor parameter-effect curves, group 2.

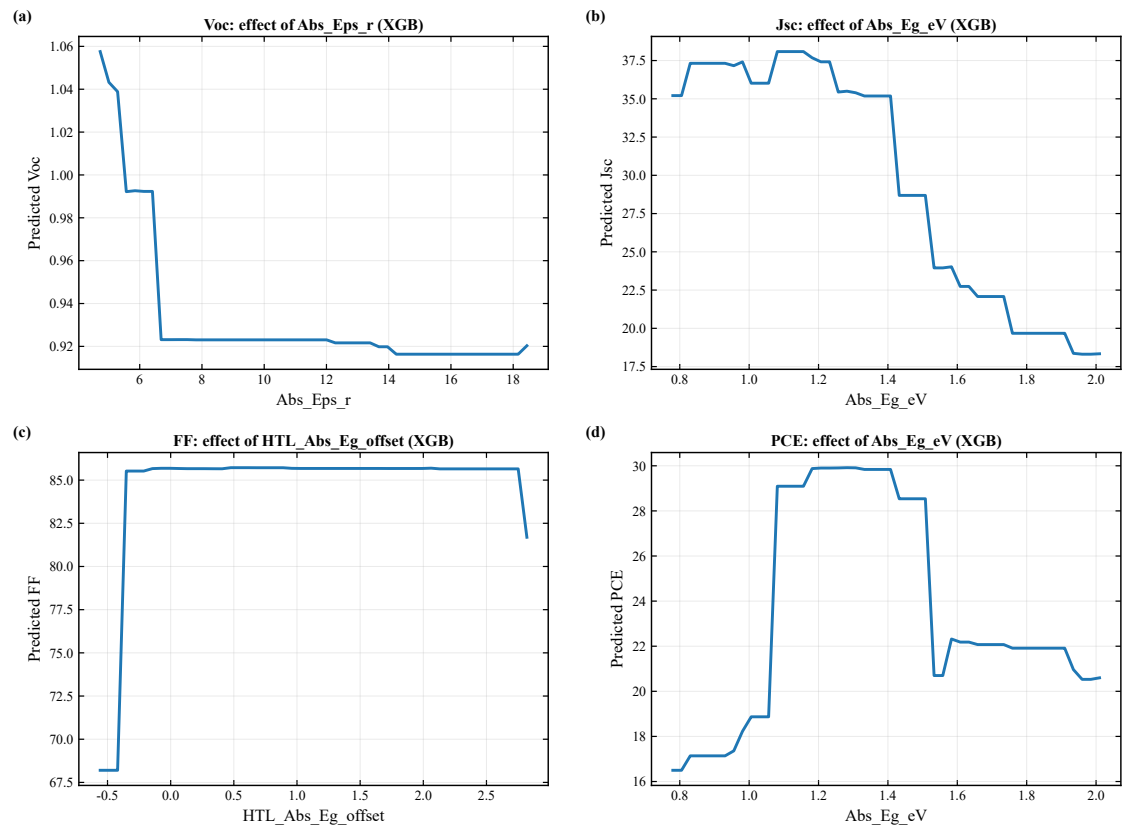


Figure S3: Additional one-factor parameter-effect curves, group 3.

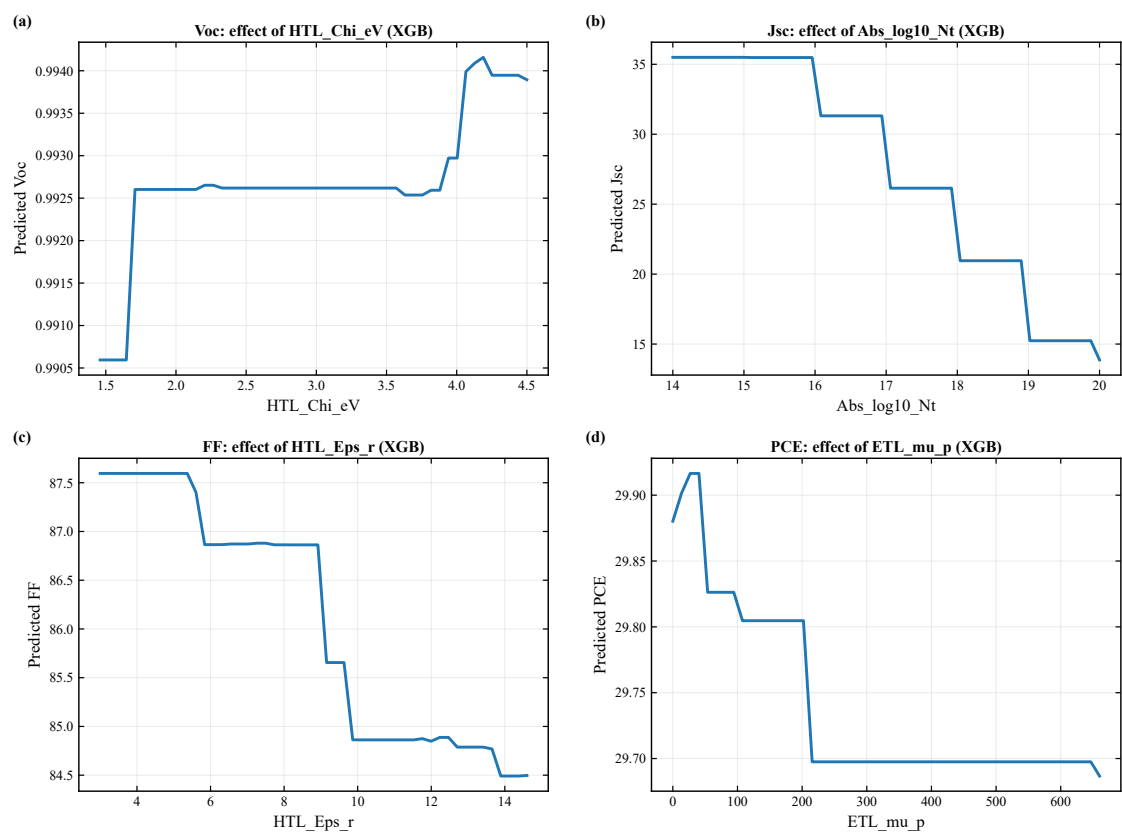


Figure S4: Additional one-factor parameter-effect curves, group 4.

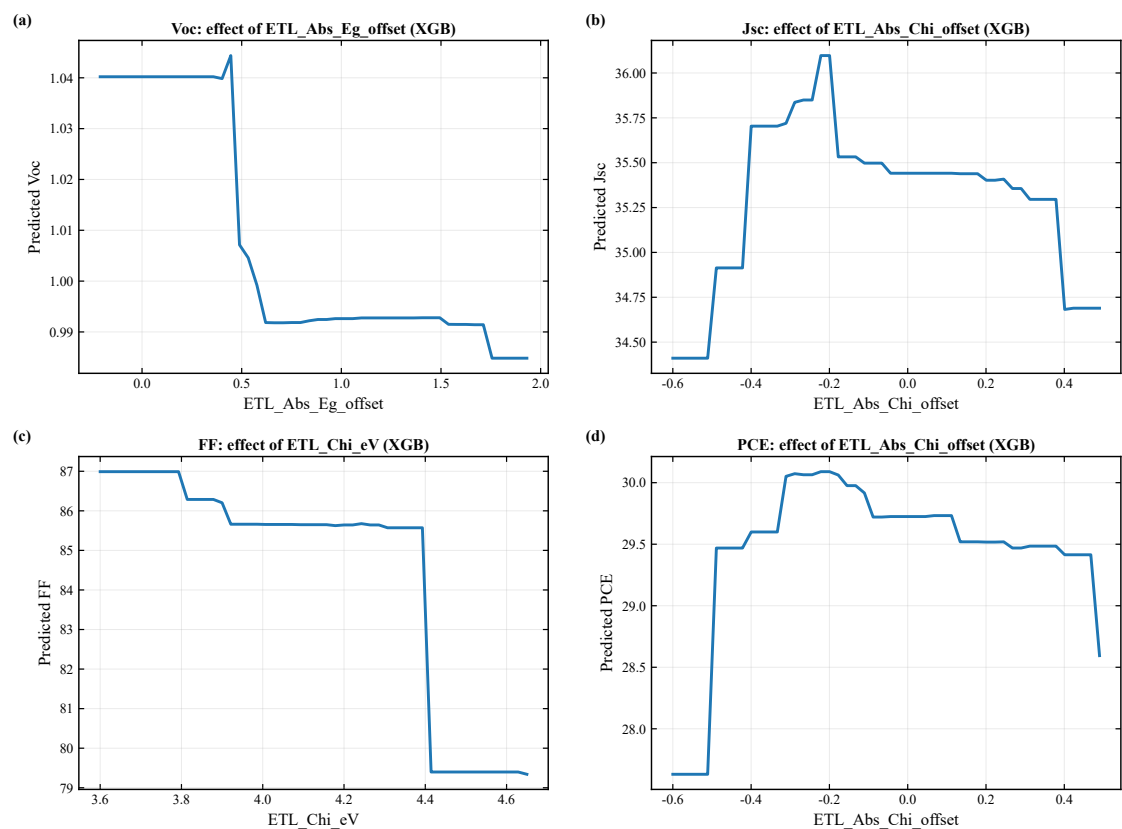


Figure S5: Additional one-factor parameter-effect curves, group 5.

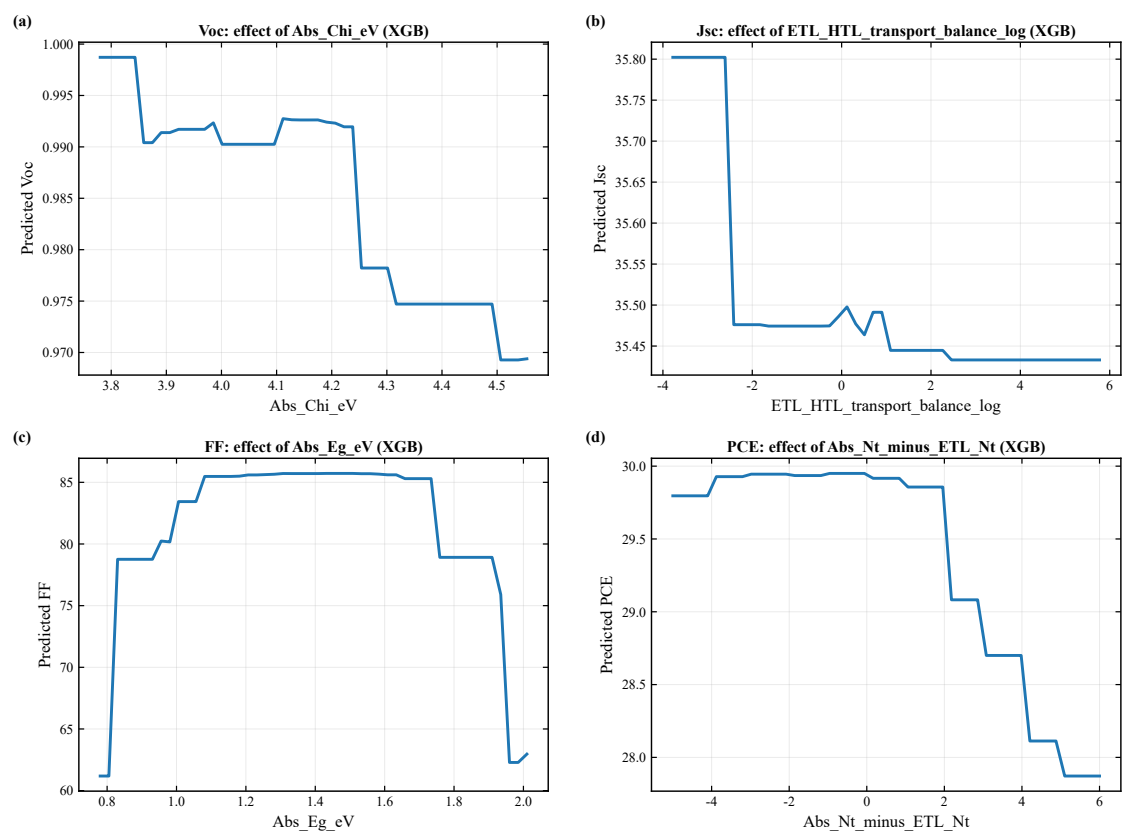


Figure S6: Additional one-factor parameter-effect curves, group 6.

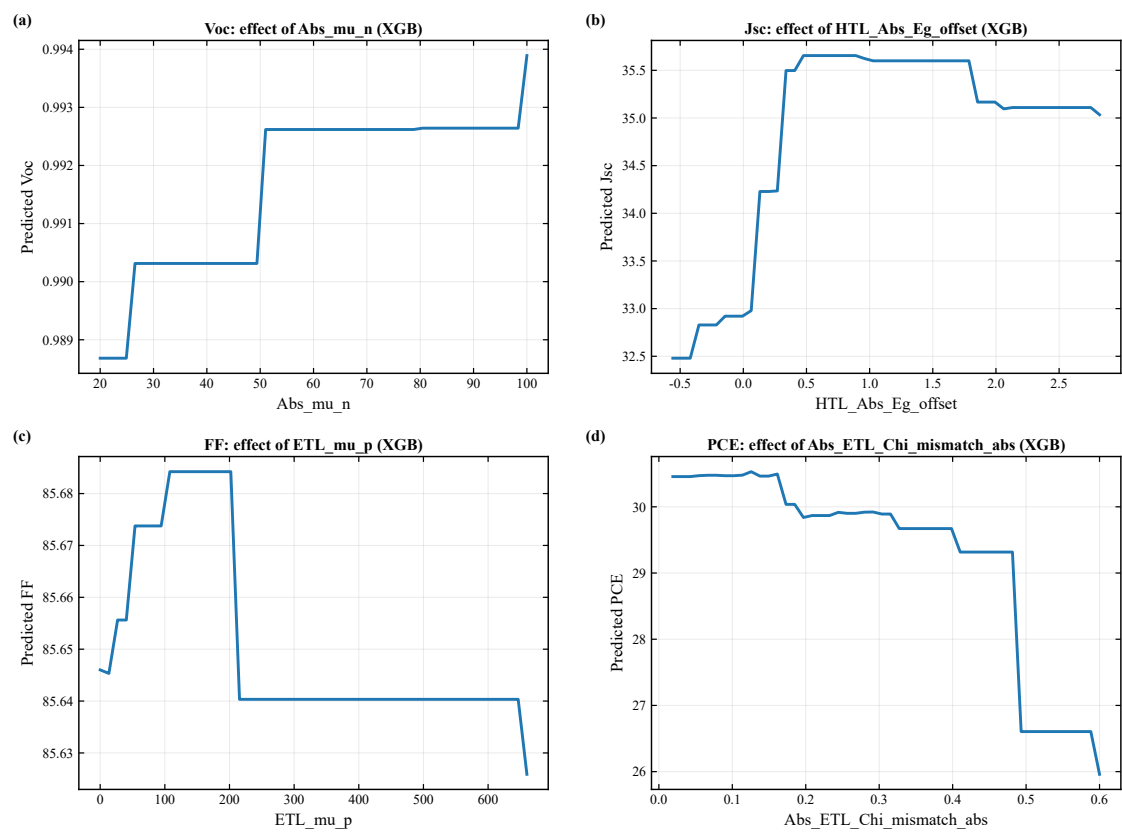


Figure S7: Additional one-factor parameter-effect curves, group 7.

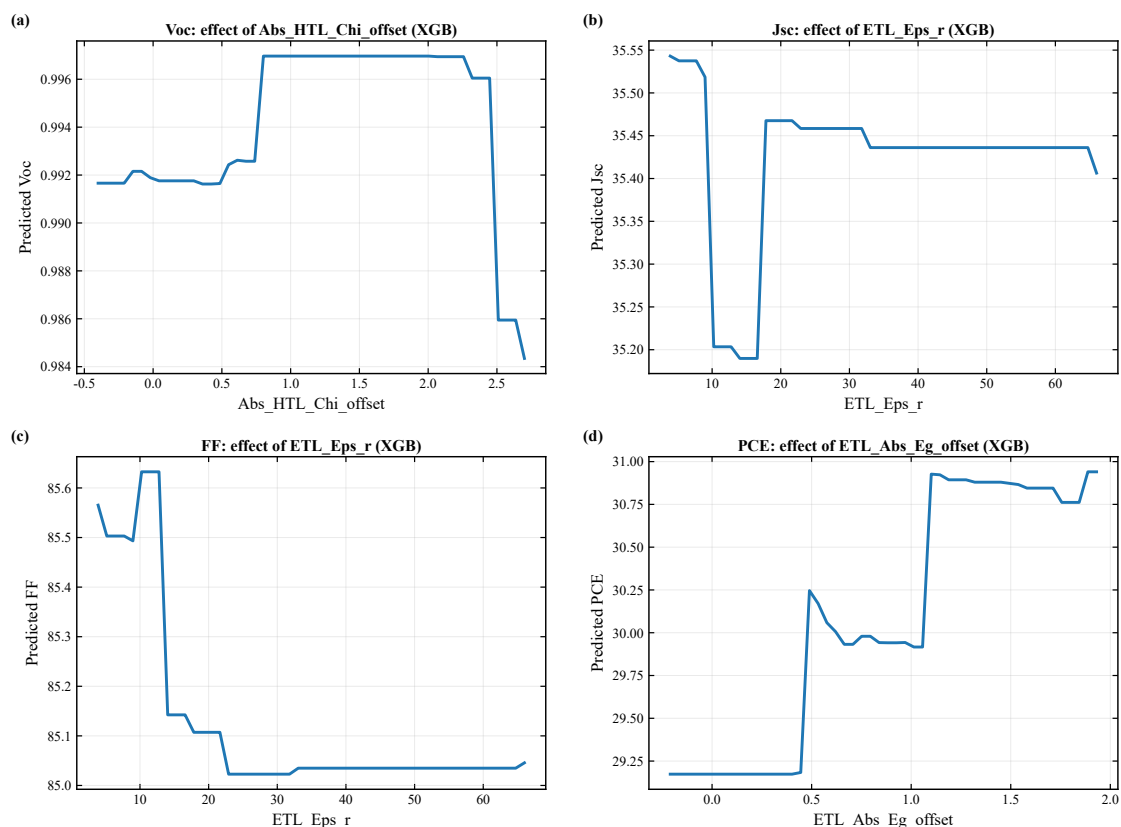


Figure S8: Additional one-factor parameter-effect curves, group 8.

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